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Insights into Malaria Data Consistency Across Service and Disease Reporting Streams in Ethiopia's HMIS/DHIS2

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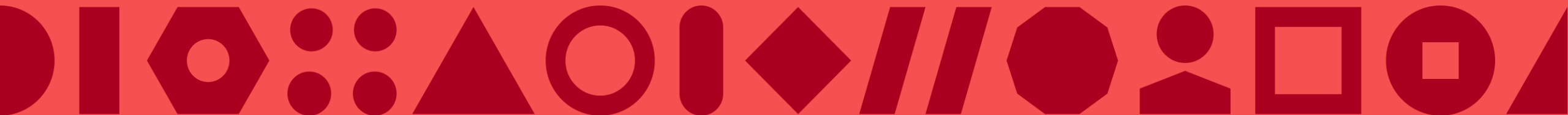
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Background



Background: Malaria in Ethiopia

- ~75% of Ethiopia is malarious; 69% of the population is at risk of malaria.
- **Malaria has resurged in Ethiopia since 2022**, reversing progress in malaria control and elimination.
- Transmission varies substantially by **geography and season**, requiring closer monitoring of where and when malaria burden is changing.
- The evolving epidemiological situation increases the need for **timely and reliable evidence** to guide program decisions and resource allocation.
- As malaria responses become increasingly **sub-nationally tailored**, reliable routine surveillance data are critical for monitoring and action.

Malaria Surveillance in Ethiopia

- Ethiopia has two major national health data systems: HMIS and PHEM.

PHEM	HMIS
• Weekly (use WHO EPI Week)	• Monthly (use Ethiopian Calendar)
• Early warning	• Routine monitoring
• Rapid response	• Trends & performance

- HMIS provides a primary evidence base for routine monitoring and management of the National Malaria Elimination Program.
- Malaria information in HMIS is captured through two reporting streams: **Service reporting** and **Disease reporting**.

HMIS provides a primary evidence base for routine malaria monitoring and program management.

HMIS Data Elements

Service	Disease
• MAL_Slides or RDT performed for malaria diagnosis • MAL_Slides or RDT Positive • MAL_Malaria with Travel History • MAL_Foci investigated and classified (Elimination Woreda) • MAL_Index cases investigated and classified (Elimination Woreda) • MAL_HHs that need LLINs • MAL_targeted HHs received at least one LLINs • MAL_Case notified to PHCU for further investigation (Elimination Districts)	ICD 10: (2018-2023)
	• B50-196 Malaria (Plasmodium falciparum malaria) • B50-197 Malaria (Plasmodium falciparum malaria with cerebral complications) • B50-198 Malaria (Other severe and complicated Plasmodium falciparum malaria) • B50-199 Malaria (Plasmodium falciparum malaria unspecified) • B50-200 Malaria (Plasmodium vivax malaria) • B50-201 Malaria (Plasmodium vivax malaria with other complications) • B50-202 Malaria (Plasmodium vivax malaria without complication) • B50-203 Mixed Malaria (Other parasitologically confirmed malaria) • B50-204 Malaria (Unspecified malaria)
	ICD 11: (2023-)
	• ESV-ICD11 1F40 - Malaria due to Plasmodium falciparum • ESV-ICD11 1F43 - Malaria due to Plasmodium ovale • ESV-ICD11 1F42 - Malaria due to Plasmodium malariae • ESV-ICD11 1F41 - Malaria due to Plasmodium vivax • ESV-ICD11 1F44 - Other parasitologically confirmed malaria • ESV-ICD11 1F40/1F41 - Malaria due to Plasmodium falciparum associated with Malaria due to Plasmodium Vivax (Mixed Malaria) • ESV-ICD11 1F45 - Malaria without parasitological confirmation

Rationale





Service and disease data are expected to tell a consistent story about confirmed malaria cases

- **Service data:** malaria tests performed and laboratory-confirmed positives, without species disaggregation.
- **Disease data:** confirmed malaria cases disaggregated by species, alongside clinically diagnosed cases.
- Because both streams originate from the same health facilities and health posts, laboratory-confirmed positives in service data would be expected to broadly align with confirmed cases in disease data.
- **Yet, routine data show a persistent discrepancy**
- The discrepancy between the two streams has not been fully characterized, leaving uncertainty about how each dataset should be interpreted and used.



The project therefore sought to quantify the discrepancy, understand its implications, and provide an evidence-based approach for selecting the appropriate HMIS variable for different analytical purposes.

Objectives



Objectives



- To quantify and characterize discrepancies between confirmed malaria positives reported through HMIS service data and confirmed cases recorded in disease data, and propose an evidence-informed approach to support NMEP in selecting and interpreting malaria data for routine analysis and reporting.

Quantify the magnitude and direction of discrepancies between service and disease data at national and regional levels, 2021–2025.

Characterize temporal, seasonal, and geographic patterns in the observed discrepancies.

Inform the application of a practical data-selection framework to guide NMEP in selecting and interpreting appropriate malaria data variables for different analytical purposes.

Product Description



Products of the Analysis



Research article

- Quantifies discrepancies between reporting streams
- Characterizes when and where discrepancies occur
- Provides evidence to support interpretation of malaria data



Decision-support pipeline

- Translates an existing MACEPA decision tree into R
- Automates reporting-stream/variable selection
- Provides a practical tool for routine NMEP analysis and reporting

Method



Method

Study design

- ✓ Retrospective observational
Secondary analysis
- ✓ **No sampling was performed**; all available routine HMIS data reported during the study period (2021–2025) were analyzed.

Data source & study period

- ✓ HMIS/DHIS2 data aggregated **at district level, with facility type retained as a dimension**
- ✓ **January 2021–December 2025**
- ✓ National and regional levels; available facility types

Data analyzed

- ✓ Service reporting vs. Disease reporting
 - **Confirmed malaria cases**
- ✓ **Clinically diagnosed cases were excluded** from the comparison.

Analysis

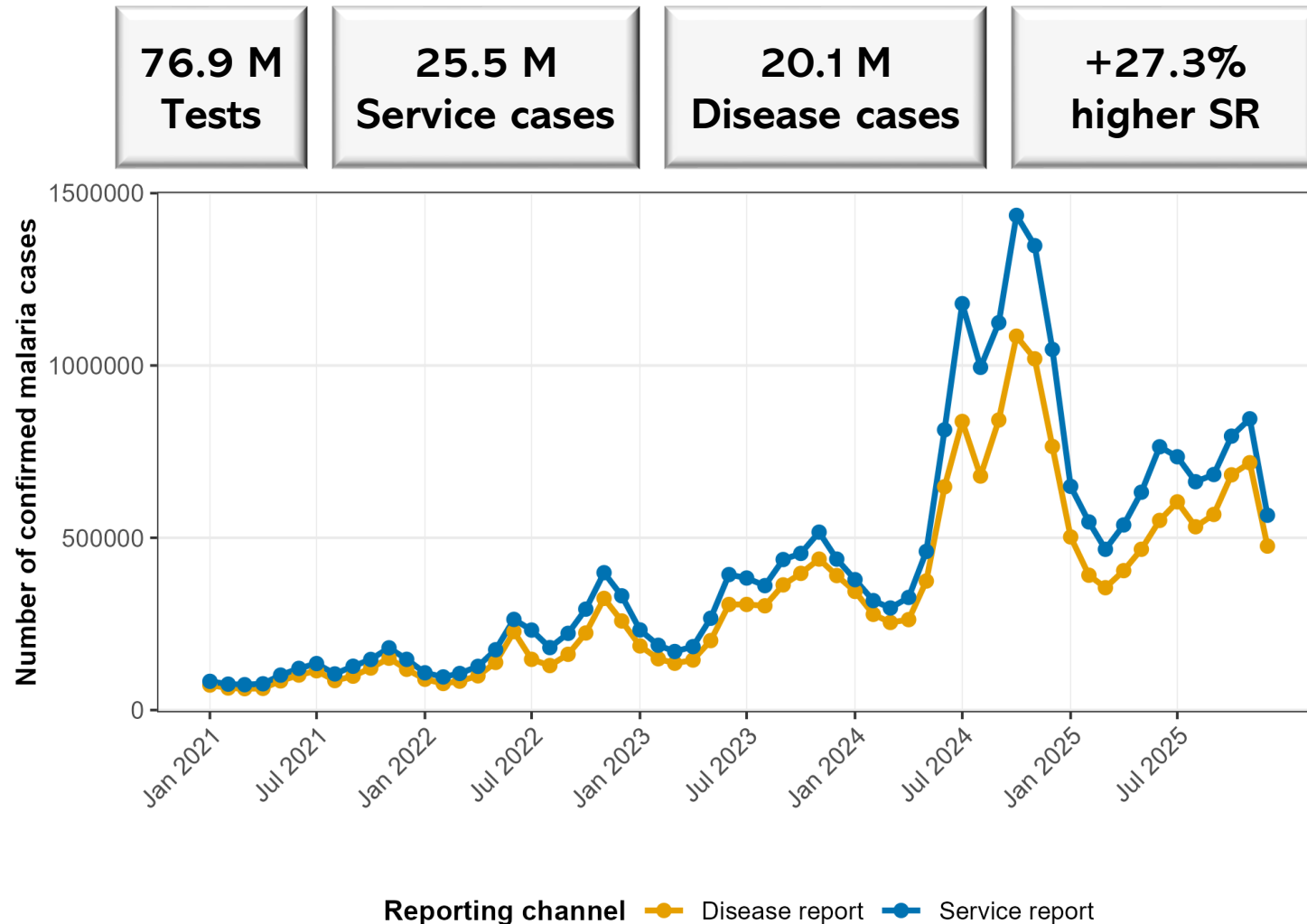
- ✓ Magnitude & direction of discrepancies
- ✓ Patterns by time, geography, and facility type
- ✓ Implications for routine data interpretation and variable selection

DHIS2 extraction → Cleaning & harmonization → Service–disease comparison → Pattern analysis → Data interpretation

Key findings



Key findings 1: Overall magnitude of discrepancy



- Both streams followed a similar overall temporal pattern
- Cases increased through 2024 and declined in 2025
- The service–disease gap was present throughout 2021–2025

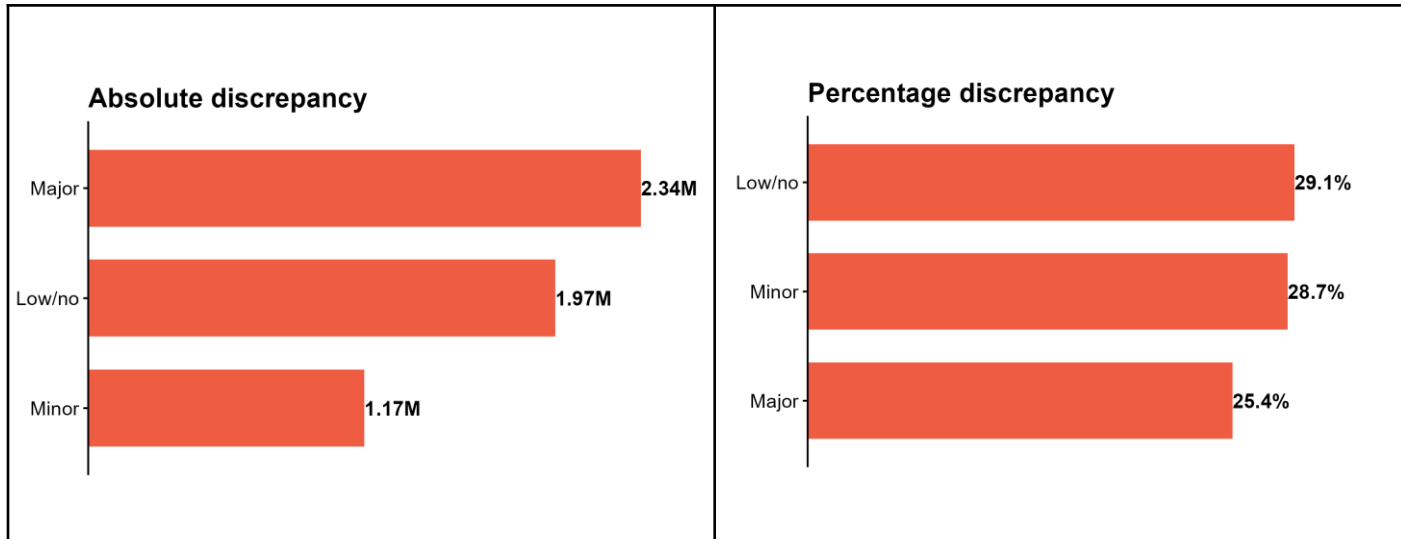
The two reporting streams showed similar temporal patterns, but service data consistently reported substantially more confirmed cases than disease data.

Key findings 2: yearly discrepancy trend

Year	Disease report (DR)	Service report (SR)	*SR higher than DR (%)	- The discrepancy was present in every year of the analysis. - It ranged from 20.8% to 31.6%. - The largest difference occurred in 2024 (31.6%). - Despite year-to-year variation, service reporting consistently exceeded disease reporting.
2021	1.14 M	1.37 M	20.8%	
2022	1.96 M	2.53 M	29.3%	
2023	3.32 M	4.02 M	21.2%	
2024	7.39 M	9.72 M	31.6%	
2025	6.25 M	7.88 M	26.1%	

The discrepancy was persistent rather than an isolated reporting anomaly, with service reporting exceeding disease reporting by approximately 21–32% each year.

Key findings 3: Discrepancies across seasons

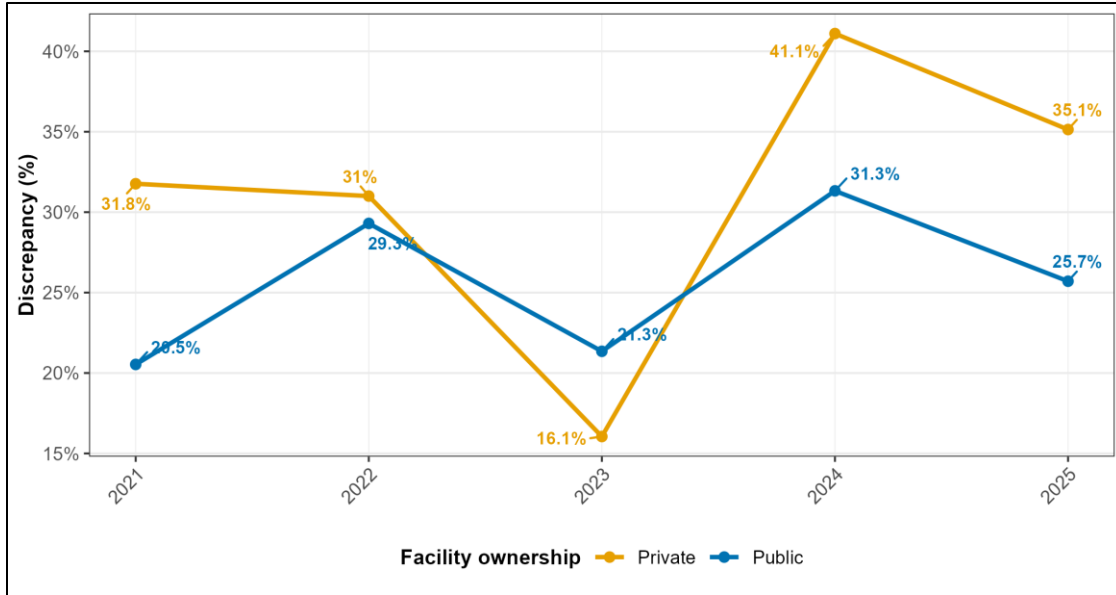


Jan–Mar + Jul–Aug → Low/no transmission
Apr–Jun → Minor transmission
Sep–Dec → Major transmission

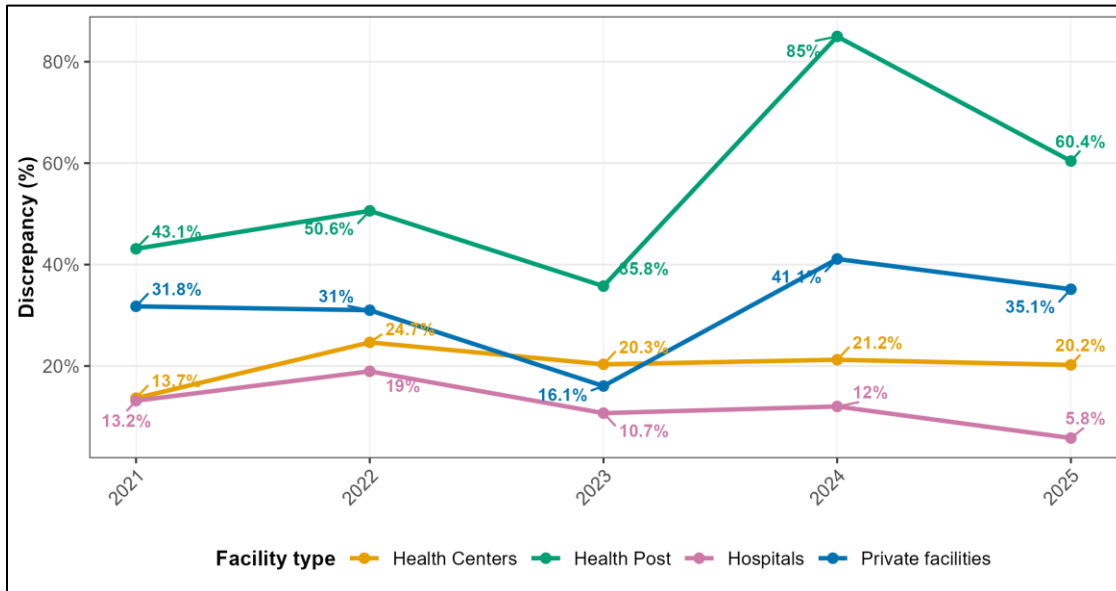
- The major transmission season had the **largest absolute gap**, reflecting higher overall malaria case volumes.
- **Relative discrepancies** were highest during the minor and low/no transmission seasons (~29%).
- The discrepancy **persisted across all transmission seasons**, indicating that it was not confined to periods of peak transmission.

The discrepancy persisted throughout the year: the major transmission season had the largest absolute gap, while relative discrepancies were greater during minor and low/no transmission seasons.

Key findings 4: Discrepancies Varied Across Facility Types

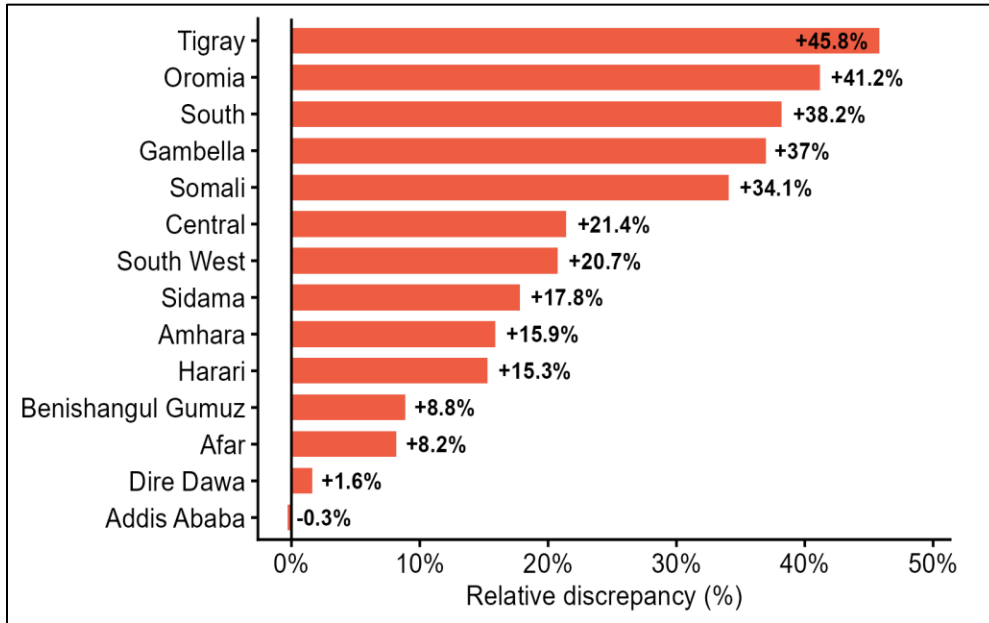


- Private facilities generally showed larger discrepancies than public facilities, but differences were more pronounced across specific facility types.
- Health posts had the largest relative discrepancy (63.7%), despite contributing only 17% of disease-reported cases & 22% of the service-reported cases.
- Health centers accounted for 68% of disease-reported cases (and 64% of the service-reported cases) and had a more moderate discrepancy (20.7%), resulting in the largest absolute difference between the two streams.
- Hospitals showed the smallest discrepancy (10.6%), while private clinics had a higher discrepancy (32.9%) but contributed only 3% of reported cases.



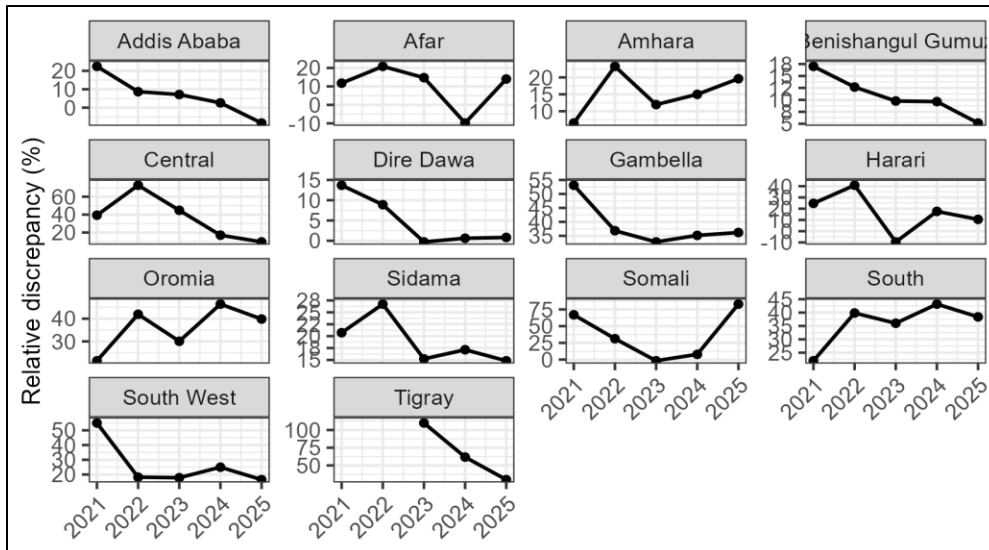
Discrepancies varied substantially by facility type: health posts had the largest relative gap, while health centers—accounting for most reported cases—contributed the largest absolute difference.

Key findings 5: Regional discrepancies



$$\text{Relative discrepancy} = (\text{service-reported} - \text{disease-reported}) / \text{disease-reported} \times 100.$$

- **Service-reported cases exceeded disease-reported cases in most regions and years**, although the magnitude and direction of discrepancies varied geographically.
- **Five-year relative discrepancies were highest in Tigray (45.8%), Oromia (41.2%), South Ethiopia (38.2%), Gambella (37.0%), and Somali (34.1%).**
- **Regional trajectories were heterogeneous:** discrepancies increased in Oromia, South Ethiopia, and Amhara, while declining overall in South West, Central, Gambella, and Tigray.
- **The direction of discrepancy was not uniformly persistent:** Addis Ababa, for example, shifted from service reports being 22.4% higher in 2021 to disease reports being 8.3% higher in 2025.
- Amhara, Oromia, SWE, and South Ethiopia- 79% of confirmed cases in both streams; but the **discrepancy didn't follow the same pattern.**



Regional differences between service and disease reporting were heterogeneous and dynamic, indicating that national-level discrepancy estimates can mask important subnational patterns

**What may explain the
observed discrepancies?**



1. Facility-level factors

- **Health posts showed the largest relative discrepancies**, potentially reflecting multiple responsibilities of health extension workers, limited data-management capacity, and weaker routine data-quality assessment.
- **Private facilities also showed substantial discrepancies**, potentially linked to limited supervision, DQA, and engagement within the evolving PPM approach.

2. Differences in recording and reporting workflows

- **Service reports are largely anchored in laboratory records**, whereas disease reports require consolidation across OPD, inpatient, emergency and other clinical registers.
- This creates **multiple points where confirmed cases may not be carried forward into disease reporting**, particularly when clinicians prioritize patient care over documentation.

3. Geographic and temporal context

- **Regional patterns were heterogeneous**, suggesting that a single national discrepancy estimate can mask important subnational differences.
- Higher workload during transmission seasons may increase absolute discrepancies, but the **lower relative discrepancy during the major season suggests that caseload alone does not explain the differences**.

Conclusion



Conclusion



Service and disease reports show **persistent discrepancies** in confirmed malaria cases across the HMIS.



The **magnitude and direction of discrepancies vary**, making a single reporting stream unsuitable for all settings.



The MACEPA-developed flowchart provides a **practical interim approach** for selecting the more appropriate reporting stream.



The findings support consideration of automated HMIS/DHIS2 validation to identify discrepancies at the point of reporting.

Improving consistency between service and disease reporting is important for reliable interpretation of malaria burden and better use of routine HMIS data

Limitations of the study



Key limitations of the study



1. Woreda-level primary analysis

- Facility-level data were unavailable for the full study period; the main analysis therefore used woreda-level data with facility type as a dimension.



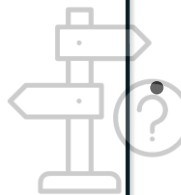
2. No external gold standard for validation

- There is no independent source against which to determine which reporting stream represents the true number of confirmed malaria cases at the facility level.



3. Pragmatic reconciliation thresholds

- The 90% and 75% thresholds provide a practical decision rule but have not been empirically validated.

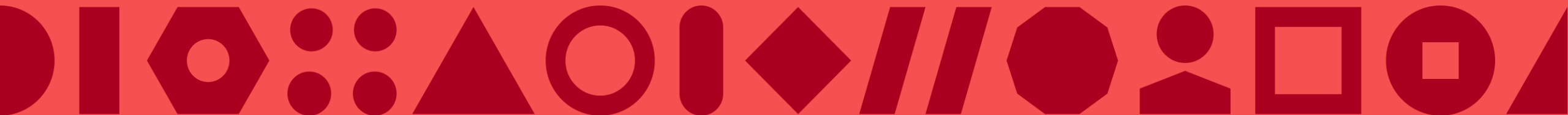


4. HMIS missingness is difficult to interpret.

- NA may represent true non-reporting or an unreported zero. The available HMIS data do not allow us to reliably distinguish between the two.

Despite these limitations, the analysis provides large-scale operational evidence on HMIS reporting discrepancies and identifies areas for improved data interpretation and future validation.

Variable Selection Flowchart: Use Case

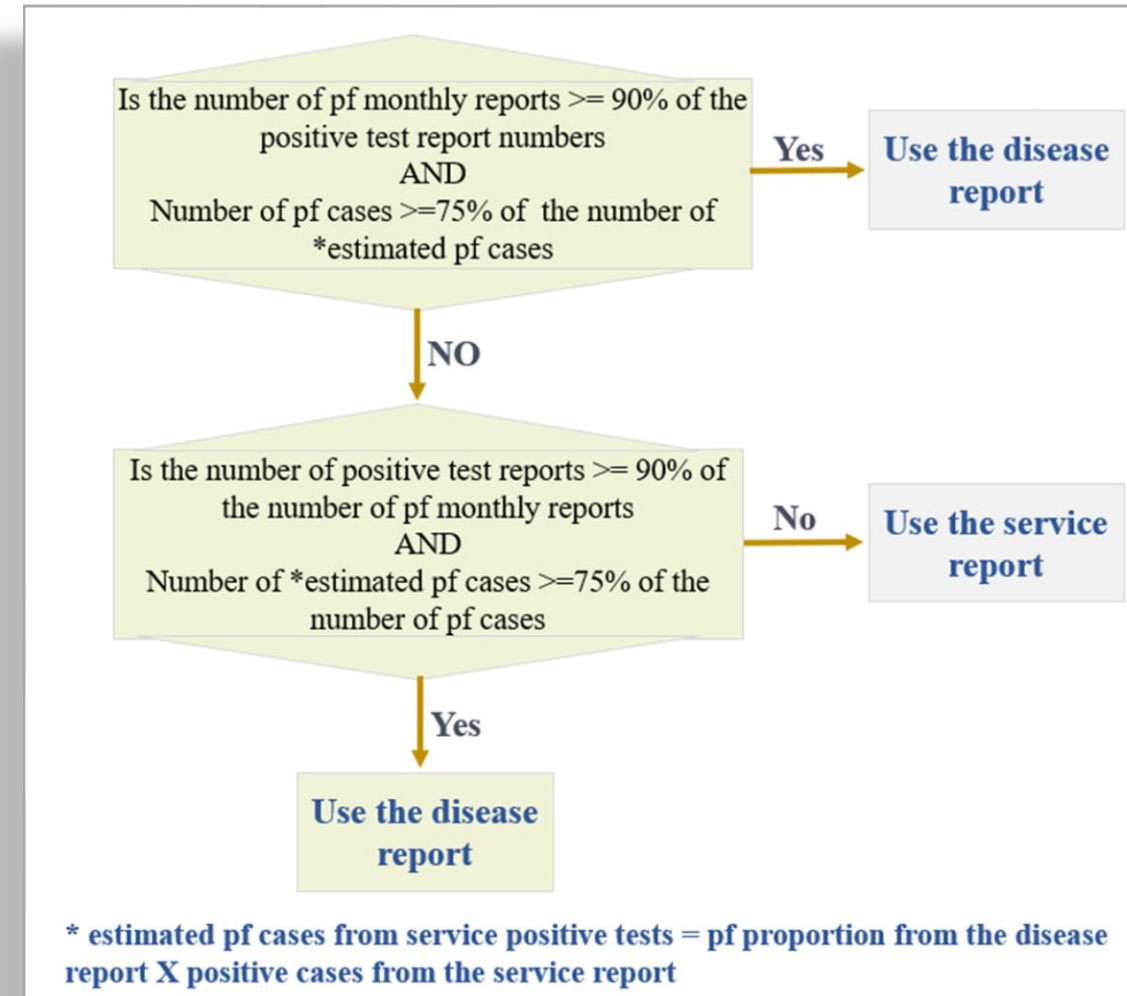


Use Case: Applying the HMIS Reporting Stream Selection Flowchart

Selected stream	n_facilities	%
Disease report	19,512	88.9%
Service report	2,430	11.1

Scenario	Confirmed cases	Difference from service report
Service report only	5,381,470	0
Disease report only	4,410,823	-970,647 (22%)
Flowchart-bases selection	5,105,154	-276,316 (5.4%)

- 2018 EFY facility-level HMIS data were used to demonstrate the application of the flowchart.
- The resulting confirmed-case estimate is compared with estimates obtained by using **either reporting stream alone**.



Benefits to the MOH



Benefits to the MOH

1. Clearer use of HMIS reporting streams

- Provides evidence on when service and disease reports diverge, helping NMEP interpret and **select the appropriate reporting stream** for specific monitoring and analytical purposes.

2. More targeted data-quality improvement

- Identifies facility types and regions with larger or persistent discrepancies, allowing DQA, supervision, and data-use efforts to be targeted rather than applied uniformly.

3. More efficient routine analysis

- Provides a standardized approach for quantifying and monitoring discrepancies, reducing repeated exploratory analysis when comparing service and disease reports.

4. Stronger confidence in malaria surveillance data

- Documents the magnitude, direction, and temporal pattern of discrepancies, providing a reference for interpreting HMIS malaria trends, reports, and analyses.

Long-term opportunity: Integrate automated DHIS2 **validation checks to flag discrepancies** between service and disease reports at the point of reporting, enabling data personnel to review and reconcile records before submission.

From identifying gaps → to targeted action → to better use of malaria surveillance data

Lessons Learned



Lessons Learned

1. Consistency across reporting streams matters

- The analysis showed that differences between service and disease reports can produce substantially different estimates of the same underlying malaria burden.

2. Reporting definitions and data flows need to be clearly aligned

- Understanding how data are generated, classified, and transferred across reporting streams is essential for identifying where discrepancies may arise.

3. Data quality directly affects programmatic estimates

- The three-scenario analysis demonstrated that the choice of reporting stream can materially change the estimated malaria burden.

4. Retrospective reconciliation has value, but prevention is preferable

- A transparent decision rule can help manage existing discrepancies, while automated validation could help identify them earlier in the reporting process.

Sustainability Plan



Sustainability Plan

1. Document the evidence

- Research report shared with NMEP as a **reference documenting the magnitude, patterns, and implications** of the service–disease discrepancy.
- Provides context for future data-quality improvement and system discussions.

2. Transfer the analytical workflow

- R-based variable-selection and reconciliation pipeline **shared with NMEP**.
- Analysts can rerun the workflow with **updated HMIS data or for different analytical purposes**.

3. Enable continued use and adaptation

- Workflow is **reproducible and adaptable**, allowing NMEP analysts to modify it as data structures and programmatic needs evolve.
- Documentation supports future users in understanding and applying the workflow.

The project leaves behind both the evidence and the analytical capability: NMEP can reuse the documented workflow to assess future HMIS data and use the findings as a reference for improving data consistency.



Sustainability Plan

